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Date: 15.11.2025

**Note:**

I used the free version of ChatGPT-4o for this assignment. To ensure that the responses were not influenced by my previous interactions or personalized model behavior, I opened a completely new browser session and used ChatGPT-4o without signing in for both prompts. Because I actively use ChatGPT Pro, the signed-in model tends to generate more tailored and higher-quality answers, even for vague prompts. Since the purpose of this assignment is to highlight the contrast between a vague prompt and an improved prompt, using an unsigned, neutral session was the most appropriate way to obtain unbiased results.

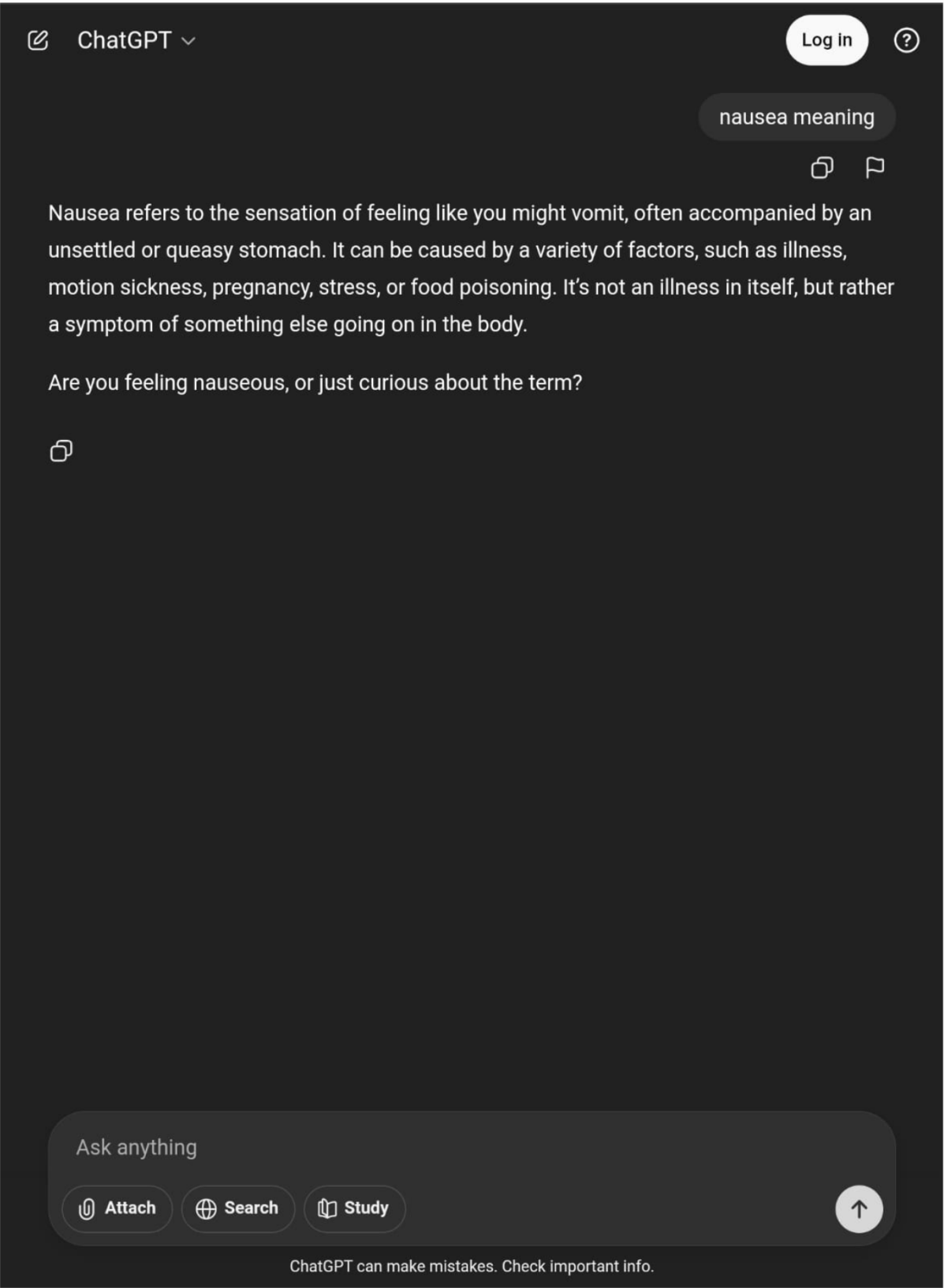
# Part 1. Prompt comparison analysis

## Domain & Task Description

This task focuses on medical patient education. I asked an AI tool to explain the symptom of nausea using two prompts: one vague and one highly structured. The goal is to demonstrate how prompt quality affects clarity, organization, and usefulness of AI-generated health information.

### a. Vague prompt

Prompt: "nausea meaning"



## B. Improved prompt

For the improved prompt, I used several prompting techniques to guide the AI's output. First, I used role prompting by asking the model to act as a family medicine doctor. I also specified the audience (a worried patient) so the tone would be calm and easy to understand.

I included a structured outline with six sections, which helped the AI organize its explanation clearly. To keep the language accessible, I added tone and vocabulary instructions, such as using simple wording and avoiding medical jargon. I also added safety constraints, asking the model not to give personalized medical advice.

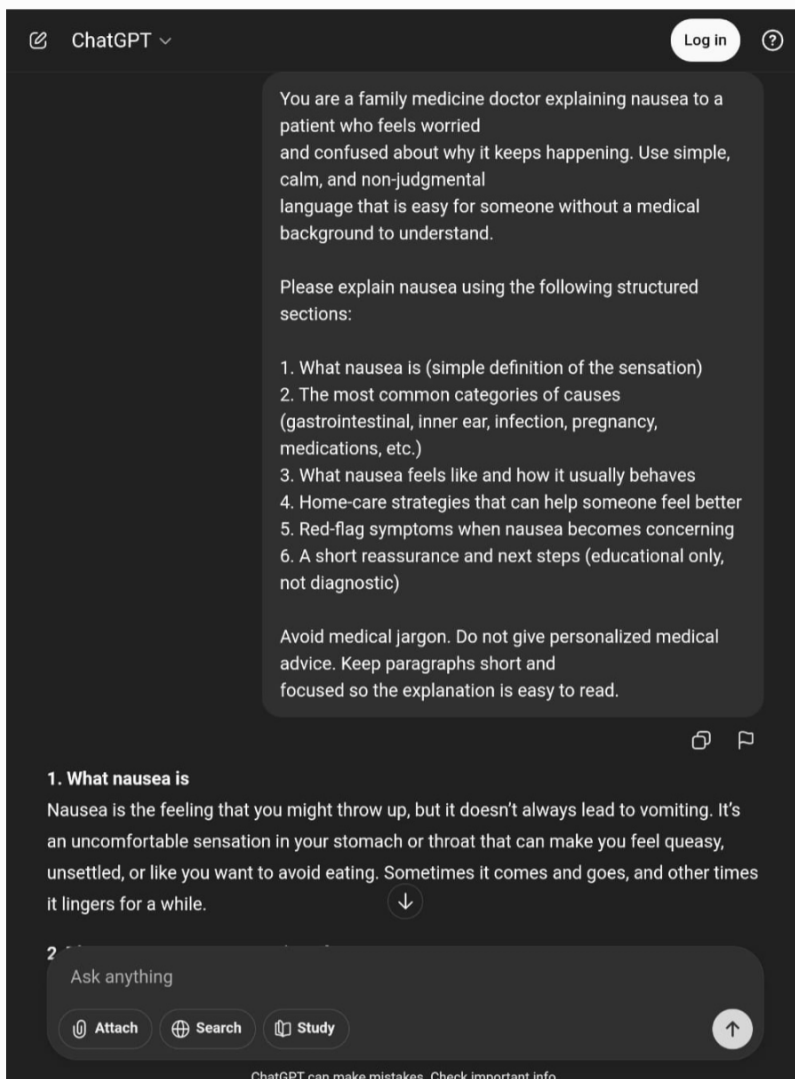
Finally, I used a brief note on length control ("keep paragraphs short"), which made the final answer easier to read. I also paid attention to the style of my own writing, because I've noticed that the AI tends to mirror the user's tone and clarity. So I made sure the prompt was written in a clean, grammatical, and respectful way to guide the AI toward that tone as well.

Prompt: You are a family medicine doctor explaining nausea to a patient who feels worried and confused about why it keeps happening. Use simple, calm, and non-judgmental language that is easy for someone without a medical background to understand.

Please explain nausea using the following structured sections:

1. What nausea is (simple definition of the sensation)
2. The most common categories of causes (gastrointestinal, inner ear, infection, pregnancy, medications, etc.)
3. What nausea feels like and how it usually behaves
4. Home-care strategies that can help someone feel better
5. Red-flag symptoms when nausea becomes concerning
6. A short reassurance and next steps (educational only, not diagnostic)

Avoid medical jargon. Do not give personalized medical advice. Keep paragraphs short and focused so the explanation is easy to read.



## 2. The most common categories of causes

Nausea can happen for many reasons, and it's often related to one of these common causes:

- **Gastrointestinal issues:** Problems like upset stomach, indigestion, or acid reflux can lead to nausea.
- **Inner ear issues:** Things like motion sickness or an infection in the ear can affect balance and cause nausea.
- **Infections:** Stomach bugs (like viruses) or other infections can cause nausea along with other symptoms like diarrhea or fever.
- **Pregnancy:** In early pregnancy, many women experience nausea, often called "morning sickness," which can happen at any time of day.
- **Medications:** Certain medicines can upset your stomach and cause nausea as a side effect.
- **Stress or anxiety:** Emotional stress can also make you feel nauseous, even if there's no physical cause.

## 3. What nausea feels like and how it usually behaves

When you're feeling nauseous, it might feel like your stomach is uneasy, queasy, or unsettled. Sometimes it can be a mild discomfort, and other times it feels stronger, making it hard to focus or do anything. It can last for just a few minutes or for hours, depending on what's causing it. Sometimes it comes and goes in waves, or you might feel it steadily.

## 4. Home-care strategies that can help someone feel better

There are a few simple things you can try at home to feel better:

- **Stay hydrated:** Sip small amounts of water or clear liquids throughout the day.
- **Eat light foods:** If you feel up to eating, try bland foods like crackers, toast, or rice.
- **Rest:** Relax and take it easy if you can. Lying down in a comfortable position may help.
- **Avoid strong smells or triggers:** Some smells, like food or perfumes, can make nausea worse.
- **Ginger or peppermint:** Some people find that ginger tea or sucking on peppermint candies can help settle the stomach.
- **Fresh air:** A short walk or just getting some fresh air can sometimes ease nausea.

## 5. Red-flag symptoms when nausea becomes concerning

While nausea is usually nothing to worry about, certain signs can indicate that it might be something more serious. Contact a healthcare provider if you experience:

- Persistent vomiting that doesn't go away.
- Severe stomach pain or cramping.
- High fever or chills.
- Blood in your vomit or stool.
- Dizziness, confusion, or fainting.
- Dehydration (feeling very thirsty, dry mouth, or not urinating).

## 6. A short reassurance and next steps

It's understandable to feel worried, but most causes of nausea are not serious and can improve with simple self-care. If your nausea continues or you have any concerns, it's a good idea to follow up with your doctor, who can help determine what's going on and guide you on the best steps to take. Keep track of when your nausea happens, what makes it better or worse, and any other symptoms you notice. This can help your doctor get a clearer picture of what's happening.

Take it one step at a time, and remember that it's okay to ask for help if you need it!



Ask anything



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## C. Analysis

When I compare the two answers, the difference is honestly pretty obvious. The first one, where I just typed “nausea meaning,” was super basic. It gave a short definition and mentioned a couple of causes, but it didn't really go anywhere beyond that. It felt like something you'd get if you quickly Googled a term. It wasn't wrong, but it wasn't very helpful either, especially if someone is actually confused about their symptoms.

The second answer was way better because I basically handed the AI a little “map” to follow. Instead of leaving it open, I told it who to act as (a family doctor) and who it was talking to (a worried patient). I also laid out the exact sections I wanted. So it wasn't guessing what to include — it had a clear outline. That alone made the output much more organised and detailed. It explained the different categories of causes properly, not just random examples. It also added red-flag symptoms and some simple home-care tips, which the vague answer completely missed.

By giving the model a role, a tone, and the exact points to cover, the explanation became way more helpful. Even the examples I included in the prompt (like GI issues, infections, pregnancy, etc.) helped guide the AI toward a fuller and more useful answer, instead of just guessing what I wanted.

## Part 2. Practical prompt design

### a. Framework selection & justification

For this task, I chose the CO-STAR framework and the business task of creating a one-month social media content calendar for a doctor/content creator. I picked CO-STAR because it gives a very clear structure to follow, which is important when you need consistent, organised content ideas rather than random posts. CO-STAR basically forces you to define the purpose, audience, tone, and the type of response you want, so the AI doesn't wander off or make assumptions. Since social media planning needs good clarity and a stable style, CO-STAR felt like the most reliable and straightforward framework for this.

### b. Complete Prompt Following Framework Structure

[CONTEXT]:

I'm helping a small business owner who's a doctor/content creator (kind of like Dr. Mike) plan out their posts. They share short, educational health content on Instagram and TikTok, and they want a proper monthly plan instead of posting randomly.

[OBJECTIVE]:  
Create a 30-day social media content calendar with clear daily ideas that can help them teach their audience and also grow their page a bit.

[STYLE]:  
Keep it simple, friendly, and evidence-based. Short sentences, easy explanations, nothing too complicated or “textbook.”

[TONE]:  
Warm, trustworthy, and approachable--like a doctor who actually cares and doesn’t talk in a scary medical way.

[AUDIENCE]:  
Regular people aged around 18–40 who scroll fast, don’t have a medical background, and prefer short, visual posts with practical tips.

[RESPONSE]:  
Please put the 30 days in a table with these columns:

- 1. Day
- 2. Topic
- 3. Post Type (reel, carousel, infographic, etc.)
- 4. A short caption idea

Try to mix different types of posts such as simple symptom explainers, myth-vs-fact content, easy health tips, red-flag signs, fun or slightly relatable doctor moments, and general lifestyle or wellness reminders. Everything should be accurate but easy to understand, since this is going on social media.

c. AI Output

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Sure! Here's a 30-day content calendar with a mix of engaging, educational, and relatable health content ideas for Instagram and TikTok. It includes a variety of post types and ideas that can help grow the page while delivering clear, evidence-based information in a warm and approachable way.

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| Day | Topic                               | Post Type   | Caption Idea                                                                                               |
|-----|-------------------------------------|-------------|------------------------------------------------------------------------------------------------------------|
| 1   | Importance of Hydration             | Reel        | "Water does so <i>much</i> for your body! 💧 Here's why staying hydrated is key to feeling good every day." |
| 2   | Myth vs. Fact: Cold weather & colds | Carousel    | "Myth: Cold air gives you a cold. ❄️ Fact: Colds are caused by viruses, not just the weather! 🦠"           |
| 3   | How to improve your sleep           | Infographic | "Struggling to sleep? Here are 5 simple tips for better rest tonight! 😴"                                   |
| 4   | Why stress affects your health      | Reel        | "Stress is a silent killer. Let's talk about how it impacts your heart, digestion, and mood."              |
| 5   | Red-flag signs of a heart attack    | Carousel    | "Know the signs: Chest pain, shortness of breath, or dizziness? 🚨 Get help fast if you feel these!"        |
| 6   | What happens during a check-up      | Story       | "Ever wonder what goes down during a check-up? Here's a quick rundown!"                                    |
| 7   | Healthy snack ideas                 | Carousel    | "Snack smarter! 🍎 These quick snacks will keep your energy up and your body happy."                        |

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|----|--------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------|
| 8  | Myth vs. Fact: Vitamin C & colds                             | Carousel      | "Myth: Vitamin C cures a cold. ❌ Fact: It might help <i>prevent</i> one, but it won't cure it!"     |
| 9  | 5-minute stretching routine                                  | Reel          | "Tight muscles? Try this 5-minute stretch to feel better FAST! 🧘"                                   |
| 10 | When to see a doctor for a cough                             | Infographic   | "Coughing for more than 2 weeks? Here's when to see a doctor. 🩺"                                    |
| 11 | Simple ways to boost your immune system                      | Carousel      | "Your immune system is your body's defender! Here are 4 ways to keep it strong year-round."         |
| 12 | My doctor moment: What <i>really</i> happens in an exam room | Reel          | "A peek behind the scenes: What I do in a routine exam. (Spoiler: It's not as scary as you think!)" |
| 13 | Benefits of walking 30 minutes a day                         | Carousel      | "No gym needed! Walking just 30 minutes a day can boost your mood, heart, and energy. 🚶"            |
| 14 | The truth about detoxes                                      | Myth vs. Fact | "Myth: You need a detox to 'reset.' Fact: Your body already detoxes on its own!"                    |
| 15 | Red-flag signs of diabetes                                   | Carousel      | "Feeling thirsty, tired, or losing weight? These could be signs of diabetes. 🚨"                     |

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|----|---------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------|
| 16 | Tips for better posture                     | Reel        | "Got a desk job? Here's a quick tip to improve your posture in 30 seconds!"                         |
| 17 | Simple breathing exercise for stress relief | Story       | "Feeling stressed? Take 5 deep breaths with me—here's a quick exercise to calm your mind."          |
| 18 | Why you should always wear sunscreen        | Carousel    | "Sun damage is real! ☀️ Here's why sunscreen should be part of your daily routine, even in winter!" |
| 19 | Easy ways to boost your energy naturally    | Infographic | "Feeling sluggish? Try these 3 simple ways to boost your energy—no caffeine required! ☕"            |
| 20 | Myth vs. Fact: Exercise & weight loss       | Carousel    | "Myth: More exercise = more weight loss. Fact: Diet and exercise work together. 🥗"                  |
| 21 | How to recognize anxiety symptoms           | Reel        | "Not just 'feeling stressed'? Here's how to spot the signs of anxiety and take action."             |
| 22 | Simple ways to reduce screen time           | Carousel    | "Too much screen time? Here are 5 easy tips to reduce it and protect your eyes 👁️."                 |
| 23 | Benefits of eating a balanced breakfast     | Infographic | "Don't skip breakfast! It can boost your energy, focus, and metabolism for the day ahead."          |

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|----|-----------------------------------------------|----------|-----------------------------------------------------------------------------------------------------|
| 24 | Red-flag signs of a stroke                    | Carousel | "A stroke is a medical emergency. Know the FAST signs: Face drooping, Arm weakness, Speech issues." |
| 25 | Doctor's advice on mental health check-ins    | Reel     | "Mental health is just as important as physical health! Here's why regular check-ins are crucial."  |
| 26 | How to prevent common injuries in the gym     | Carousel | "Stay safe while working out! Here's how to avoid the most common gym injuries. 💪"                  |
| 27 | Myth vs. Fact: Antibiotics & viral infections | Carousel | "Myth: Antibiotics cure all infections. ❌ Fact: Antibiotics only work on bacterial infections!"     |
| 28 | How much sleep do you really need?            | Reel     | "Not all sleep is created equal! Here's how much you really need for peak health."                  |
| 29 | Benefits of laughter on your health           | Carousel | "Laughter really is the best medicine! Here's how it boosts your mood, heart, and immune system."   |
| 30 | Quick relaxation tips for busy people         | Story    | "Feeling overwhelmed? Try these 3 quick ways to relax, even on your busiest days!"                  |

This calendar mixes practical health advice with fun, relatable content. You can adjust the topics and types of posts based on the audience's engagement or any specific goals the business owner may have for the month!

Ask anything

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For this task, I used the CO-STAR prompt I wrote, and the AI produced a full 30-day social media content calendar. The output came in a table exactly the way I asked, with columns for day, topic, post type, and caption ideas. It also followed the content style I described, meaning the posts are short, friendly, and quite easy to understand for the general public.

I highlighted a few places where the AI clearly followed the framework, f.e., the context part of the framework was followed because all the calendar topics fit the niche of a doctor/content creator. The objective was also met, since the AI actually produced all 30 posts and separated them clearly. In terms of style and tone, the captions sound warm, supportive, and not overly medical, which is exactly what I asked for. The AI also kept the audience in mind by using simple explanations and visual-friendly formats like reels and carousels. Finally, it respected the requested response format by presenting everything in the exact table layout I specified.

You can see that the AI stuck to the instructions pretty closely, even the part about mixing different types of posts. The result looks consistent with the goal of helping a small medical content creator plan a full month of posts.

#### **d. Refinement and iteration**

When I looked at the first calendar, it was honestly good, but I still felt like something was a bit “samey.” Some of the topics were repeating too closely (like too many myth-vs-facts in a row), and the flow of post types didn’t feel super natural. Also, a few captions sounded a bit plain, like they were just explaining instead of actually grabbing attention. So I wanted the second version to feel more dynamic.

To fix this, I wrote a clearer follow-up prompt:

“Please revise the calendar by increasing topic variety (fewer repeated formats), adding stronger engagement hooks in the captions, and making sure the post types alternate more naturally. Keep the same table format.”

The updated output was definitely better. The topics became more balanced, so it didn’t feel repetitive anymore. The captions also changed a lot—now they start with stronger hooks like “Did you know?” or “Tap through for the real facts,” which makes them more engaging for social media. And the post types alternated smoothly instead of being clumped together.

Overall, the refined prompt worked way better because I actually pointed out what was wrong and gave the AI a clear direction, instead of hoping it would guess what I meant.

#### **e. Framework effectiveness reflection**

Using the CO-STAR framework actually made the whole prompt way easier to structure. I didn’t have to guess what to include—the framework basically forced me to think through each part: who the content is for, what the exact objective is, the tone, the style, and the final response format. If I hadn’t used a framework, my prompt probably would’ve been too vague, and the output wouldn’t have been nearly as organised or useful.

For this specific task, I think CO-STAR was honestly the best option, because it balances both creative and practical parts. I don’t think RACI or SMART would’ve fit as well—they’re more for responsibilities or measurable outcomes, not something more creative like social media planning.

What I learned is that choosing the right framework really matters. If you pick one that doesn’t match the task, the prompt feels awkward. But when the framework matches the task, everything comes out way clearer and more “intentional.”

### **Part 3. Critical Reflection**

#### **a. Skill progression**

I’ve been working with AI tools since I was quite young, experimenting with early beta models long before they became mainstream. Because of that, I’ve spent years observing how small changes in tone, structure, or style can completely shift the output. Over time, I also noticed something interesting: the more consistently I interacted with a model, the more it seemed to “learn” my preferences. Even vague prompts started giving surprisingly accurate answers because the AI already understood the level of clarity and detail I tend to expect. I’ve even created a few personal AI agents tailored to

specific tasks, so I didn't enter this course as a beginner.

What this course has given me, though, is a more academic and structured perspective. I finally learned how to classify and articulate the strategies I had been using intuitively for years. Seeing my habits reflected in formal frameworks-like CO-STAR-was genuinely an aha moment. It helped me organize my approach with much more intention.

#### **b. Practical application**

In the future, I can see myself using prompt engineering in my medical career, especially in areas like clinical decision support, research, or communicating complex scientific information in a clearer way. Medicine involves huge amounts of data, and being able to guide an AI system to summarize articles, compare treatment guidelines, or generate structured reports would save a lot of time. Frameworks like CO-STAR or SMART would be useful because they force me to define the depth, tone, and structure I need, which is essential when accuracy actually matters.

One challenge I expect is that medical AI must be handled cautiously-models can sound confident even when they're wrong. Another challenge is preventing information oversimplification, because patients and colleagues need clarity, not shortcuts. To deal with that, I'd use AI only as a supportive tool: I'd refine prompts carefully, verify outputs against reliable sources, and keep the final interpretation in human hands. Prompt engineering would simply help me work more efficiently and more deliberately in a future clinical or research environment.

#### **c. Limitations and Ethics**

Two major limitations I'd have to keep in mind when using AI in the future are accuracy and overconfidence. Large language models can phrase things in a very convincing way, even when parts of the information are incomplete or subtly wrong. They also tend to generalize, which can be risky in any medical or scientific setting where small details matter. Another issue is that AI sometimes "fills in the gaps," which can create false confidence if I don't double-check what it produces.

An important ethical consideration is making sure AI never replaces critical judgment, especially in medical contexts where decisions can affect real people. To use AI responsibly, I would keep prompts transparent, avoid asking for anything diagnostic, verify all outputs with trusted sources, and treat the model as a support tool instead of an authority. Clear prompting and careful verification would help me stay on the safe side.

#### **d. Goals for the second half**

In the second half of the course, I want to explore more advanced prompting strategies, especially techniques that improve precision and reduce ambiguity in complex tasks. I'd also like to get better at choosing the right framework instantly. My main goal is closing the gap between "intuitive prompting" and fully intentional, expert-level prompting.